

## Quality Assurance and Management

# Product management assignment

### Project description

InnoScream – Anonymous Stress Relief Platform for Students

### Project Goal

Develop a Telegram-based platform where students can anonymously share their frustrations (e.g., deadlines, exams) and receive community support via reactions, memes, and analytics. Designed for 1 week of development by a team of 5.

### Core Features

1. Anonymous Posts:
  - Command `/scream [text]` to post anonymously. Example: `/scream 9 AM lectures should be illegal 🤡`
  - Reactions (🤡 / 🔥 / 😄) to upvote posts.
2. Top Screams:
  - One per day show daily most-voted screams.
  - Auto-generates memes from top posts (ImgFlip API or *alternative*).
3. Analytics:
  - Weekly stress graphs (QuickChart.io or *alternative*).
  - `/stats` – Personal post count (anonymous tracking by `user_id`).
4. Moderation:
  - Admins can delete posts via `/delete [id]`.

### Technical Stack

- Backend: Python 3.11 + FastAPI + SQLite.
- Telegram Bot: *aiogram* (or *alternative*).
- APIs:
  - Memes: ImgFlip (or *alternative*).
  - Charts: QuickChart.io (or *alternative*).
- Quality Tools: pytest, Flake8, Bandit, Locust.

## Quality Requirements

### Maintainability

- Flake8 with zero warnings (code style).
- Docstrings for all functions (enforced via pydocstyle).
- Code complexity  $\leq 10$  (measured with radon cc).

### Reliability

- Test coverage  $\geq 60\%$  (pytest-cov).
- Mutation testing  $\geq 80\%$  survival rate (mutmut).

### Performance

- Bot response time  $\leq 500$  ms under 100 RPS (Locust).
- SQL queries  $\leq 50$  ms (verified via EXPLAIN ANALYZE in SQLite).

### Security

- No critical vulnerabilities (Bandit).
- Data anonymity: user\_id hashing (audited via Bandit/manual review).

## CI/CD Integration

Automated quality gates via GitHub Actions (or *alternative*)

## Negotiation Process

The Product Manager (PM) and the development team will collaborate to finalize quality targets during the initial project sign-off phase. These discussions will focus on adjusting measurable thresholds, such as increasing test coverage to 70% if the team confirms feasibility. All discussions should be completed within one week of project assignment to ensure timely progress. The final agreement will be documented in the README of the project repository or in a dedicated quality manifesto for transparency.

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